Nairobi Securities Exchange

NSE Stock Risk Intelligence System

Machine Learning-Driven Behavioral Risk Clustering

57+ Stocks Analyzed | 26 Risk Indicators | ML-Powered

2024

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Chapter 01

Project Overview

Transforming raw market data into actionable risk intelligence

Project Overview

The Problem: Information Asymmetry

Herding Behavior

Despite the Nairobi Securities Exchange seeing a significant surge in equity turnover, many retail investors in Kenya suffer from "herding behavior" —relying on social intuition rather than technical data to make investment decisions.

Information Gap

The lack of accessible, data-driven risk assessment tools leaves everyday investors vulnerable to market volatility and poor decision-making, creating a barrier to informed participation in Kenya's growing equity market.

The Opportunity

This project bridges the information gap by converting raw historical stock price data into behavioral risk clusters using machine learning, empowering investors with evidence-based insights.

The Current Reality

Investors Using Data	15%
Investors Using Intuition	85%

Source: Behavioral finance research on emerging market retail investors

Main Objective

To develop a data-driven stock market intelligence system for the NSE that analyzes historical price behavior and sector characteristics to support informed, risk-aware investment decisions.

Specific Objectives

- Engineer financial and time-series features including returns, volatility, liquidity, and momentum
- Analyze sector-level performance and risk characteristics across NSE-listed companies
- Apply unsupervised machine learning to segment stocks by behavioral patterns
- Identify and classify high-risk, stable, and speculative stocks within the NSE
- Deliver insights through a clean, interactive Streamlit dashboard accessible to non-technical users

Project Overview

The Solution: Risk-Based Segmentation

System Architecture

The system groups 57+ major NSE stocks into risk-based categories using 26 engineered features , enabling investors to move from intuition to evidence-based decision-making.

Feature Engineering

26 distinct metrics including volatility, risk, technical indicators, and liquidity measures

ML Clustering

K-Means and DBSCAN algorithms segment stocks into distinct risk profiles

Interactive Dashboard

Streamlit-based interface for real-time exploration and risk assessment

Risk Categories

91%	6%	3%
Low Risk	Moderate	High Risk

Interactive Dashboard Preview

Streamlit-powered interface for real-time stock exploration

Chapter 02

Methodology

From raw data to intelligent risk profiling

Methodology

Feature Engineering: 26 Risk Indicators

Volatility Metrics

Mean Volatility	Price fluctuation average
5-Day Volatility	Short-term variance
Downside Deviation	Negative return risk

Risk Indicators

Maximum Drawdown	Peak-to-trough decline
VaR (95%)	Value at Risk metric
Sharpe Ratio	Risk-adjusted return

Technical Signals

RSI	Relative Strength Index
MACD	Momentum indicator
Bollinger Band Width	Volatility bands

Liquidity Measures

Amihud Illiquidity	Price impact measure
Trading Frequency	Activity level
Volume Metrics	Daily turnover

Plus 14 additional engineered features for comprehensive risk profiling

Methodology

Machine Learning Pipeline

Clustering Algorithms

K-Means Clustering

Primary

Primary model for segmenting stocks into Low Risk, Moderate Risk, and High Risk categories. Optimal k=4 determined through elbow method and silhouette analysis.

Hierarchical Clustering

Validated

Produced identical results to K-Means, confirming cluster stability and robustness of the segmentation approach.

DBSCAN

Retired

Left 54% of stocks unclassified due to density-based limitations. Not suitable for this dataset's distribution characteristics.

Dimensionality Reduction

PCA (Principal Component Analysis) used for visualizing high-dimensional stock clusters in 2D/3D space, enabling intuitive interpretation of risk groupings.

Model Performance

0.72

Silhouette Score

Excellent cluster separation

K-Means	0.72
Hierarchical	0.72
DBSCAN	0.28

Silhouette score ranges from -1 to 1, where higher values indicate better-defined clusters

Chapter 03

Key Insights

Data-driven discoveries from the NSE analysis

Key Insights

Cluster Analysis: Risk Distribution

0 Cluster 0: Low Risk

91% of NSE Stocks

Moderate volatility with muted returns, reflecting the characteristics of a low-liquidity emerging market bourse.

Volatility	Low-Moderate
Returns	Muted

Note: Risk label ≠ return quality. Several Cluster 0 stocks (SGL, TCL, UCHM) have negative mean returns.

1 Cluster 1: High Momentum

IMH, KPLC, KEGN

Shows the strongest 60-day momentum in the universe but also the deepest historical drawdowns—high-frequency trend plays with significant tail risk.

60-Day Momentum	+37.7%
Max Drawdown	-55.2%

Strategy: Tactical exposure with strict position caps (≤10%)

2 Cluster 2: Outlier

ORCH

Statistical outlier with extraordinary 60-day return but near-zero trading frequency—a liquidity mirage, not a replicable signal.

60-Day Return	+205%
Trading Frequency	4.25%

Warning: Illiquidity makes this cluster unsuitable for most investors

3 Cluster 3: Avoid

KNRE

The only stock with a negative Sharpe ratio, negative mean return, and daily trading—the clearest "avoid" signal in the universe.

Sharpe Ratio	Negative
Mean Return	Negative

Recommendation: Clear avoid signal for all investor types

Key Insights

Sector-Level Intelligence

Banking Sector Leadership

The Banking sector leads quality within Cluster 0, demonstrating a rare convergence of capital preservation and near-term price strength on the NSE.

SCBK **+33.2%**

Standard Chartered Bank Kenya

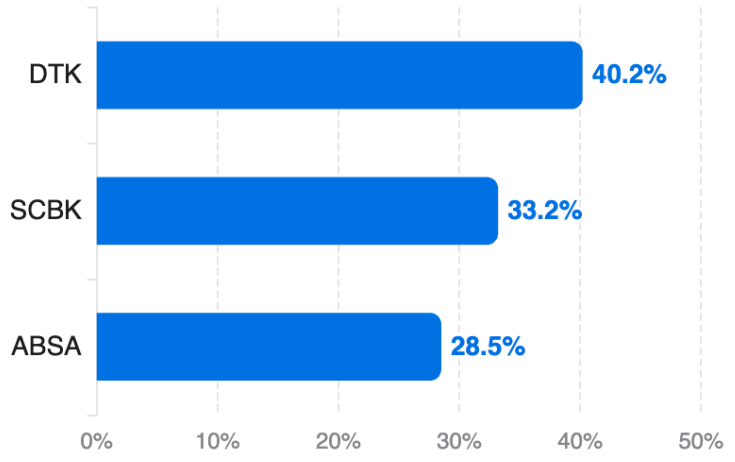
DTK **+40.2%**

Diamond Trust Bank

ABSA **+28.5%**

ABSA Bank Kenya

Banking Sector Performance



60-day momentum performance of top banking stocks in Cluster 0

Key Takeaway

These banking stocks combine low volatility with the strongest 60-day momentum in the low-risk tier—an optimal balance for conservative investors seeking growth.

Chapter 04

Recommendations

Strategic guidance for investors and stakeholders

Recommendations

Investment Strategy Framework

Low-Risk Portfolio Picks

Best selections from Cluster 0 with confirmed upward momentum—ideal for conservative investors seeking stable returns.

SCBK +33.2%
Standard Chartered

DTK +40.2%
Diamond Trust Bank

ABSA +28.5%
ABSA Bank Kenya

COOP Stable
Co-operative Bank

NCBA Stable
NCBA Group

Important Note

Risk label does not equal return quality. Several Cluster 0 stocks (SGL, TCL, UCHM) have negative mean returns. Always verify individual stock performance within clusters.

Tactical/Growth Exposure

Cluster 1 stocks with strict position caps (≤10%)

IMH Momentum Play
I&M Holdings

KPLC High Volatility
Kenya Power

KEGN Growth Potential
KenGen

⚠ These stocks show strong momentum but carry significant tail risk. Limit exposure to ≤10% of portfolio.

Recommendations

Implementation Roadmap

1

Build Investor App

Create a simple website or mobile app where any investor can search for an NSE stock and immediately see risk level and performance metrics.

Keep it accessible for non-finance backgrounds

2

Educate Investors

Hold workshops and create videos explaining what different risk categories mean and how to use them for better stock selection.

Short videos and simple graphics

3

Promote Best Picks

Highlight Cluster 0 stocks with confirmed upward momentum: SCBK, DTK, ABSA, COOP, NCBA as low-risk recommendations.

Banking sector leadership

4

Data-Driven Decisions

Push investors to stop following the crowd and look at real numbers. Show key indicators in everyday language.

Move from intuition to evidence

5

Tactical Guidelines

Establish position caps for high-risk exposure: Cluster 1 stocks (IMH, KPLC, KEGN) limited to ≤10% of portfolio.

Risk management protocols

Expected Impact

By implementing this roadmap, retail investors will transition from herd-driven decision-making to evidence-based investing, reducing exposure to high-risk assets while capturing growth opportunities in Kenya's equity market.

Technology Stack & Future Enhancements

Current Technology Stack

Languages
Python

Data Processing
Pandas, NumPy

Visualization
Matplotlib, Seaborn, Plotly

ML Framework
Scikit-learn, PCA

Web App
Streamlit

Deployment
Streamlit Cloud

Development
Jupyter Notebook

Version Control
Git, GitHub

Future Enhancements

Real-Time Feeds
Incorporate real-time NSE price feeds for live risk assessment and dynamic portfolio monitoring.

Sentiment Analysis
Add sentiment analysis from financial news and social media to enhance risk predictions.

Macro Integration
Extend clustering to include macroeconomic indicators for broader market context.

Portfolio Optimization
Implement portfolio optimization algorithms based on risk clusters for automated allocation.

Conclusion

From Intuition to Evidence

A practical, data-driven tool that helps everyday investors make better decisions—moving away from following the crowd and toward evidence-based investing in Kenya's stock market.

57+	26	0.72
Stocks Analyzed	Risk Indicators	Silhouette Score

Data-Driven ML-Powered Investor-Focused